



The Relationship between a Business and its Information Needs

From data to wisdom — how a business gathers data, turns it into information, uses it in its roles, and protects it from threats

TIME	SESSION	MODULE	QUALITY ASSURANCE
90 minutes · Self & Group	Friday, 28 Aug 2026 · 09h00 – 14h00	Module 1 · Professional Team Development	QCTO / MICT SETA

Ackoff's five categories of the human mind

According to Russell Ackoff, systems theorist and professor of organisational change:



Data

Symbols — raw facts with no significance beyond their existence.



Information

Data processed to be useful — answers who, what, where, when.



Knowledge

Application of data and information — answers how.



Understanding

Appreciation of why — carries each level to the next.



Wisdom

Evaluated understanding — vision and design.

PAST AND FUTURE

The first four categories deal with the past — what has been or what is known. Only wisdom deals with the future; people must move successively through the other categories to reach it.

Data and information



Data is raw

It simply exists — usable or not — and has no meaning of itself. In computer terms, a spreadsheet generally starts out by holding data.



Information has meaning

Data given meaning by relational connection. In computer terms, a relational database makes information from the data stored within it.

THE WEATHER EXAMPLE

Data: “It is raining.” · Information: “The temperature dropped 15 degrees and then it started raining” — a relationship, possibly cause and effect.

Knowledge, understanding and wisdom



Knowledge

The appropriate collection of information — memorised, deterministic. The times table answers 2×2 , but not 1267×300 .



Understanding

Cognitive and analytical — synthesises NEW knowledge from what is known. Learning, not memorising. AI systems (2026: Copilot, Gemini) do this at scale.



Wisdom

Extrapolative, non-deterministic. Calls on moral and ethical codes; judges right from wrong. The authors: computers will never possess it.

THE WEATHER EXAMPLE, CONTINUED

Knowledge: high humidity + a sharp temperature drop usually means rain. Wisdom is systemic: “it rains because it rains” — grasping all the interactions between evaporation, air currents and temperature gradients.

When is a pattern knowledge — and when is it noise?

- “Abugt dbesbt regtc uatn s uitrzt...” — 100% novelty. To you it is noise; there is no foundation to connect with the pattern. Yet translated, these are Newton's three laws of motion.
- Is something knowledge if you can't understand it?
- Now: a heavy 3×3×6 box with a door, food inside, colder inside than out, usually in the kitchen, ice compartment, light when it opens, dirt underneath, junk on top...
- A refrigerator — at some point you connected with the pattern, and every further statement only confirmed it. In a society without refrigerators, it stays a riddle.

THE POINT FOR BUSINESS

Information only becomes knowledge when the receiver can connect it to a pattern they understand — present information in the language and context of its users, or it is noise.

From information to a final choice

- Decision making is the cognitive process leading to the selection of a course of action among variations — every process produces a final choice: an action or an opinion.
- It begins when we need to do something but know not what — a reasoning process, rational or irrational, on explicit or tacit assumptions.
- Structured, rational decisions drive science-based professions — e.g. medical diagnosis, then treatment selection.
- Under time pressure, high stakes or ambiguity, experts decide intuitively: recognition-primed decisions fit the indicators into experience and arrive immediately at a satisfactory course of action.
- Robust-decision approaches formally integrate uncertainty into the process.

Price control



Free price system

Prices set by supply and demand — signals between producers and consumers that ration supplies, distribute income and allocate resources.



Controlled price system

Prices set by government within a controlled market or planned economy.

IN THE ORGANISATION

Your prices are largely governed by what competitors charge for similar products. Pricing, costing and manufacturing must communicate so goods sell at the same or better price than competitors.

USING THE INFORMATION

Quality control and PDCA

Quality systems ensure products and services meet or exceed customer requirements — “fit for purpose”, “do it right the first time”. QA covers design through documentation.



Plan

Decide the objective and the process to deliver it.



Do

Carry out the plan; collect data along the way.



Check

Measure results against the expected outcome.



Act

Correct differences; standardise what worked.

COMMUNICATION

Competitors' items are checked and measured against the organisation's own quality — that comparison depends on communication.

Marketing and business performance



Marketing

R&D meets sales & marketing so new products are fully understood, then marketed to consumers.
Two-way: client surveys flow back to R&D to build what customers request.



Business performance (BPM)

Processes that organise, automate and analyse the metrics, processes and systems driving performance — the next generation of business intelligence (BI).

GROWTH THROUGH COMMUNICATION

Analyse the performing areas, understand the tools that work, and apply them where the business performs less well. All views, opinions and input are needed for educated decisions.

Three threats you must be able to explain



Unauthorised access

Hackers or insiders reading data they have no right to see. Defend: passwords + MFA, per-role access rights, encryption — POPIA makes it a legal duty.



Viruses & malware

Software that corrupts, steals or destroys data — in 2026 above all ransomware via phishing email. Defend: antivirus, patches, backups, cautious staff.



Disgruntled staff

Insiders who deliberately damage, leak or delete data. Defend: least-privilege access, audit trails, remove access the day someone leaves.

The information needs of the sub-systems

Sub-system	Information it needs
HR (employment)	Employee records, contracts, leave and attendance, payslips, performance reviews
Production (manufacture)	Orders, raw-material stock, production schedules, quality measurements
Marketing (branding)	Brand assets, campaign results, customer feedback and survey data
Financial (cash flow)	Invoices, payments, debtors and creditors, cash-flow forecasts, budgets

Your work for Lesson 4



Questioning session

90 minutes · Self & Group — all 14 questions: data, information, the five roles, the three threats and the four sub-systems. AI-marked.



Slide quizzes

Every section gates on its five questions — all correct unlocks Next.



Self assessment

“I am able to explain the relationship between a business and its data requirements” — tick it honestly, or write it down as a goal.



Logbook project — Research

Your IT-in-business research project also evidences this outcome. Mark it 114050.



Data is raw. Information has meaning. Wisdom stays human.

Data → information → knowledge → understanding → wisdom · decisions, price control, quality (PDCA), marketing and BPM · guard it against unauthorised access, malware and insiders · give HR, Production, Marketing and Finance the information they need.